MODULE 4

The Laser Beam and the Echo

4.1 The beam spreads out

SIMPLE APPROACH

Shine a flashlight at your hand: you see a small circle. Now shine it at a wall across the room: the circle is much bigger. A laser beam spreads too — just very, very slowly. Far away, the laser "dot" can be as big as a coin, so it can touch the edge of a pole and the wall behind it at the same time.

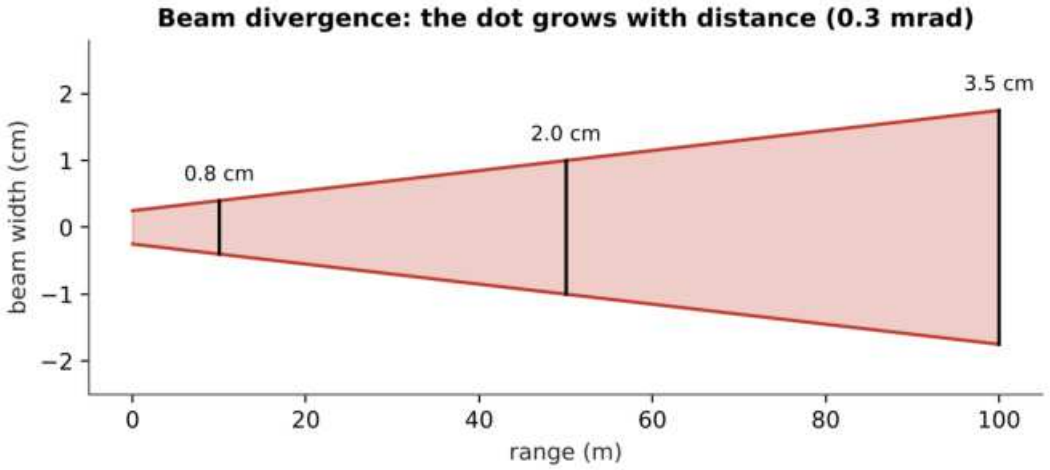


Figure 4.1: The footprint of a beam with 5 mm exit diameter and 0.3 mrad divergence.

KEY EQUATION — Beam footprint

$$D(\rho) \approx d_0 + \rho\,\gamma$$

What every symbol means

Symbol	Meaning	Units
D	diameter of the laser spot on the target	m
d_0	diameter of the beam as it leaves the sensor	m
ρ	range	m
γ	full-angle beam divergence (small-angle rule again!)	rad (often given in mrad = 10^{-3} rad)

WORKED EXAMPLE — spot size at 50 m and 100 m

Given: $d_0 = 5$ mm $= 0.005$ m, $\gamma = 0.3$ mrad $= 0.0003$ rad.

At 50 m

$D = 0.005 + 50 \times 0.0003 = 0.005 + 0.015 = 0.020$ m $= 2.0$ cm

At 100 m

$D = 0.005 + 100 \times 0.0003 = 0.035$ m $= 3.5$ cm

Result: a 2–3.5 cm "blur" at mapping ranges. You cannot locate an edge more sharply than about half the spot size.

4.2 How strong is the echo? The LiDAR range equation

SIMPLE APPROACH

The farther away something is, the dimmer its echo. It gets dim fast: twice as far means only one quarter as bright. Shiny white things (a road sign) send back lots of light; black asphalt or wet surfaces send back very little. And if the light hits a surface at a slant, most of it bounces away like a ball skipping off water.

KEY EQUATION — Received power for an extended diffuse target

$$P_r = P_t \frac{\rho_{\text{refl}} \cos\theta_i}{\pi R^2} A_r \eta_{\text{sys}} \eta_{\text{atm}}^2$$

What every symbol means

Symbol	Meaning	Units
P_r	power arriving at the detector	W
P_t	power transmitted in the pulse	W
ρ_{refl}	target reflectance (0 = black, 1 = perfect white)	—
θ_i	incidence angle between beam and surface normal	rad
R	range	m
A_r	area of the receiving lens	m^2
η_{sys}	optical efficiency of the sensor	—
η_{atm}	one-way atmospheric transmission (squared: out and back)	—
π	comes from a diffuse (Lambertian) surface scattering into a half-sphere	—

WORKED EXAMPLE — comparing two targets

Given: Target A: white concrete ($\rho_{\text{refl}} = 0.6$) at 10 m, head-on. Target B: asphalt ($\rho_{\text{refl}} = 0.1$) at 50 m, hit at 60°. Everything else equal.

Step 1 — ratio

$$\frac{P_B}{P_A} = \frac{0.1}{0.6} \times \frac{\cos 60^\circ}{\cos 0^\circ} \times \frac{10^2}{50^2}$$

Step 2

$= 0.1667 \times 0.5 \times 0.04 = 0.00333$

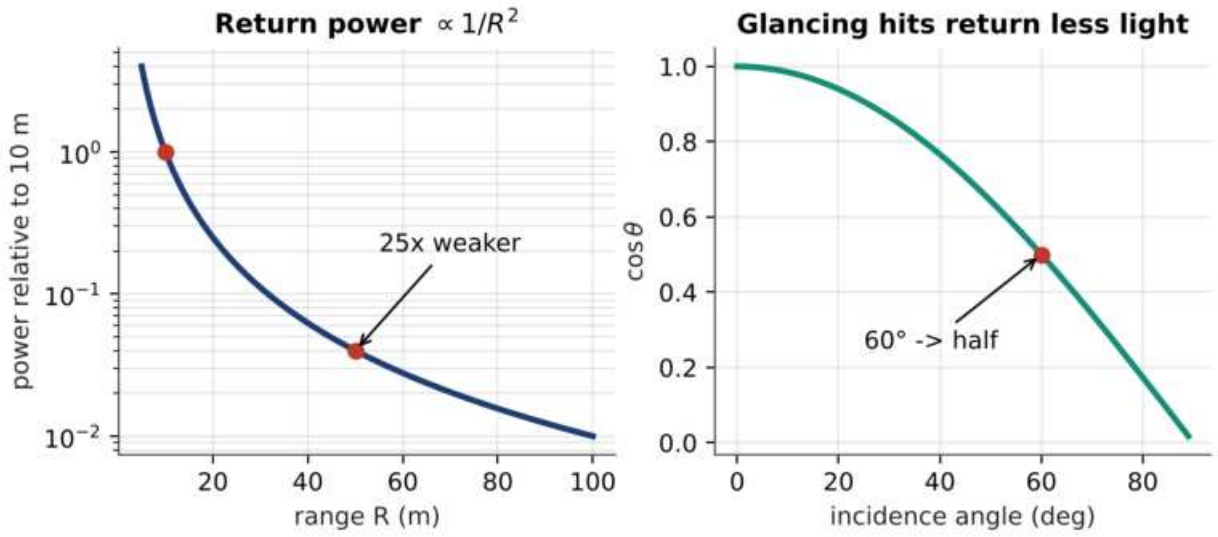


Figure 4.2: Left: the $1/R^2$ law. Right: the $\cos\theta_i$ incidence-angle law.

Result: B returns about 1/300 of A's power. Weak echoes have noisier ranges, which is why range precision gets worse with distance and grazing angle — and why road surfaces far from the vehicle are the noisiest part of a mobile scan.

4.3 Intensity

Every point carries an **intensity** value, a number related to P_r. Because of the R^2 and $\cos\theta_i$ effects, raw intensity of the same paint changes with range and angle; software can *normalize* it. Intensity makes road markings, signs and retro-reflective targets jump out of the cloud, which is handy for extracting line work and for finding control targets.

4.4 Wavelength and eye safety

ENGINEER'S VIEW

Most automotive and mobile scanners use **905 nm** (near-infrared, cheap silicon detectors) or **1550 nm** (absorbed by the eye's cornea and lens before reaching the retina, so far more power may be used eye-safely — giving longer range). All survey scanners used in public must be **Class 1** eye-safe under IEC 60825-1. Water strongly absorbs 1550 nm, so wet surfaces and rain reduce returns more at that wavelength; green (532 nm) lasers are used for bathymetric LiDAR because green light penetrates water.

4.5 What can LiDAR see through?

SIMPLE APPROACH

A laser is light, so it acts like a flashlight beam, not like X-ray vision. It cannot see *through* a wall, a tree trunk, or the ground. It *can* slip through holes — the gaps between leaves, the openings in a fence — and reach whatever is behind them. It passes through clear glass the way sunlight does. And it is stopped almost right away by water.

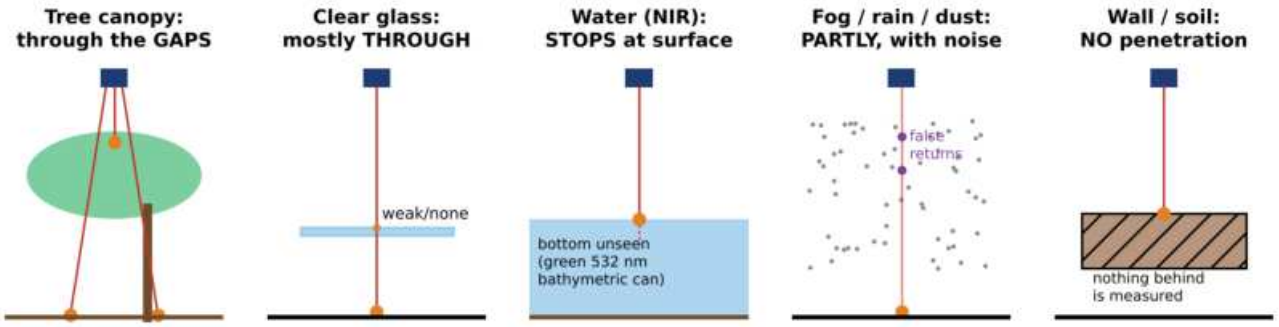


Figure 4.3: The five situations every LiDAR user meets. Orange dots are recorded returns.

Material or condition	Does the laser get through?	What you get and what to do
Tree canopy, shrubs	**Through the gaps only**	First returns on leaves, last returns on the ground where there are openings. Dense evergreen or summer canopy may leave no ground hits: collect in leaf-off season, slow down, add passes.
Tall grass, crops, mangroves	**Rarely**	Points sit on top of the vegetation, not the soil. Heights are biased high; take ground shots by conventional methods.
Chain-link fence, mesh, railings	**Partly** (gaps)	Some points on the fence, some behind it; the fence appears as a sparse ghost surface.
Clear glass windows	**Mostly through**	Few returns from the glass itself (except head-on); you see the room behind. Reflections can create "mirror-world" ghost points.
Tinted, frosted or coated glass	**Partly / varies**	Unpredictable; check the cloud.
Mirrors, polished metal, chrome	**No — reflects away**	Dropouts or ghost points placed at false ranges behind the surface.
Water (865–1550 nm)	**No**	Absorbed within centimeters; calm water often returns nothing except straight down. Only green (532 nm) bathymetric LiDAR maps the bottom, in clear water.
Wet pavement, puddles	**No**	Weak or missing returns; specular reflection; schedule collection after surfaces dry.
Rain, fog, dust, smoke, steam	**Partly**	Range shortens; "floating" false points near the sensor; heavy fog can block completely. Filter them out (Module 18).
Falling snow	**Partly**	Noise points in the air; snow cover hides the ground surface.
Walls, soil, pavement, people, vehicles	**No**	Opaque: anything behind is in shadow. Plan extra passes or positions to see behind objects.
Underground utilities	**No**	LiDAR maps only visible surfaces (manhole lids, valve covers). Use ground-penetrating radar or locating for what is buried.

BIG IDEA

LiDAR never "penetrates" solid material. It reaches hidden ground only through openings, which is why multiple returns (Module 3) and many viewing angles matter more than laser power.

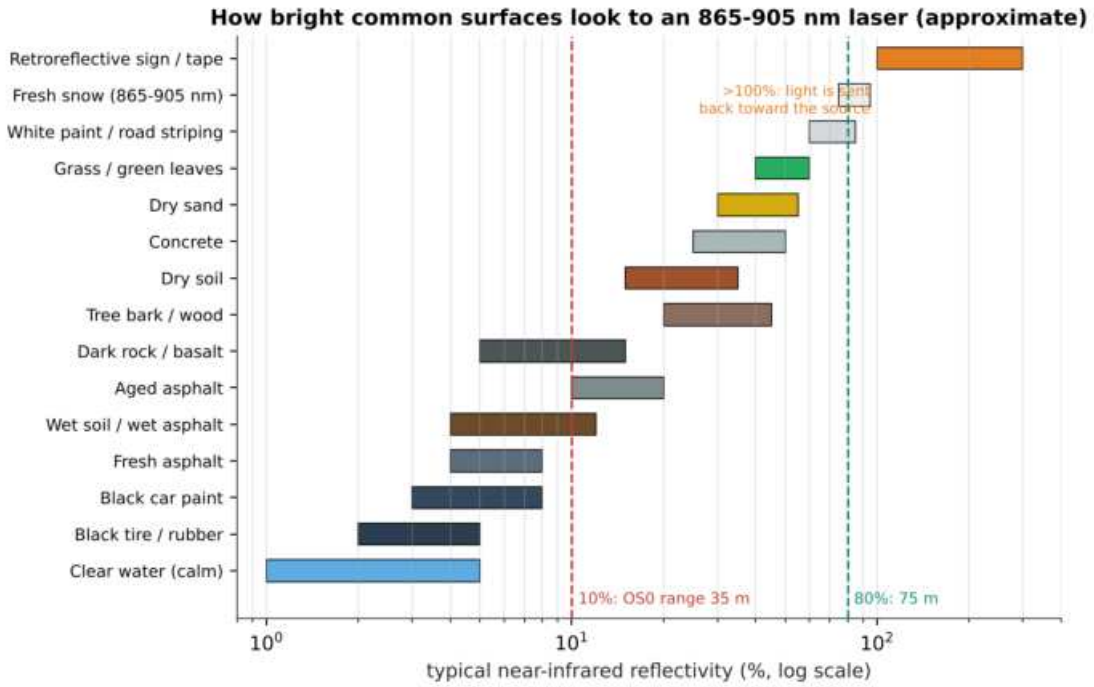


Figure 4.4: Typical reflectivity of common surfaces at the near-infrared wavelengths used by most mobile scanners (865–905 nm). Values vary with moisture, age, roughness, color and incidence angle.

4.6 Reflectivity of common surfaces

Surface	Typical reflectivity	What it means in the field
Clear, calm water	1–5%	often no return at all except straight below the sensor
Black tire, rubber	2–5%	among the darkest targets; shortest detection range
Black or dark car paint	3–8%	cars may appear with holes; glossy paint can also mirror
Fresh asphalt	4–8%	dark: road returns thin out first at long range
Wet soil, wet asphalt	4–12%	water darkens surfaces in the near infrared
Dark rock (basalt)	5–15%	dark
Aged, gray asphalt	10–20%	brighter as it weathers
Dry soil	15–35%	medium
Tree bark, wood	20–45%	medium
Concrete	25–50%	good, consistent target
Dry sand	30–55%	bright
Grass, green leaves	40–60%	bright in the near infrared (healthy plants reflect strongly there)
White paint, road striping	60–85%	very bright: striping stands out in intensity images
Fresh snow	75–95% at 865–905 nm	very bright at 905 nm — but much darker at 1550 nm
Retroreflective signs and tape	“over 100%”	sends light straight back; can saturate the receiver and bias the range (Module 15)

Values are typical ranges compiled for teaching; they are not specifications. Sensors that report *calibrated reflectivity* (such

as the Ouster OS0) let you measure your own surfaces.

WORKED EXAMPLE — how far can I see dark asphalt?

Given: the OS0 detects a 10% target to 35 m (Module 15). Received power falls as $1/R^2$ (Section 4.2), so for the same detection threshold the maximum range grows as the *square root* of reflectivity.

Fresh asphalt, 5%
$35 \times \sqrt{5/10} = 35 \times 0.71 = 25$ m

Black car, 4%
$35 \times \sqrt{0.4} = 22$ m

Concrete, 40%
$35 \times \sqrt{4} = 70$ m (the datasheet's 75 m at 80% shows the rule flattening out in bright sunlight)

Result: plan the usable swath on the *darkest* surface you must map: on new asphalt, expect road coverage to end near 25 m, long before the brochure range.

4.7 Range precision in practice

Situation	Typical range noise (1σ)
Good mobile unit, bright hard target, under 30 m (100 ft)	1–3 cm (0.03–0.10 ft)
Same unit, dark or grazing target, 50–100 m (165–330 ft)	5–15 cm (0.16–0.49 ft)
Survey-grade mobile mapping head	3–8 mm (0.01–0.03 ft)
Terrestrial phase scanner, short range	1–3 mm (0.003–0.010 ft)

These are order-of-magnitude illustrations; always read the specification sheet and, more importantly, *test on your own control*.

TRY IT YOURSELF

4.1 A beam has $d_0 = 8$ mm and $\gamma = 0.5$ mrad. How wide is the spot at 40 m?
4.2 By what factor does echo power drop when a target moves from 20 m to 60 m?
4.3 (Simple) Which gives a brighter echo: a white sign or a black tire? Why?
4.4 (Simple) Which of these can a mobile LiDAR measure the ground behind: a leafy tree, a brick wall, a glass storefront, a pond?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," of the Reference Book, starting on page 163.

SOURCES FOR THIS MODULE

Baltsavias (1999), *ISPRS J.* 54 (range equation, footprint); IEC 60825-1:2014 (laser classes); Ouster OS0 Rev8 datasheet (calibrated reflectivity, detection ranges at 10% and 80%); reflectivity values are typical teaching ranges — measured spectra are available in the USGS Spectral Library Version 7 (Kokaly *et al.*, 2017).